MODEL CARD

Mediapipe Multiclass Segmentation 15/03/2023

MODEL DETAILS

Two models to predict segmentation masks for human subjects in real time from images captured by the human subject. Runs via Mediapipe image segmenter API for on-device in real-time. The model is a multiclass segmentation model and classifies each pixel as [background, hair, body, face, clothes, others]. Supports single or multiple people in the frame, selfies and full body images.



Left: Input frame. Middle: Predicted category mask.

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MODEL SPECIFICATIONS

Model Type

- Vision Transformer Neural Network

Model Architecture

- Vision Transformer, with customized bottleneck and decoder architecture for real-time performance.

Inputs

- normalized RGB image, at resolution of [256, 256, 3] for smaller model and [512, 512, 3] for larger model. The input does not need to be resized to match the image size, and can take portrait, or landscape or any other aspect ratio images as input.

Output(s)

- Segmentation mask, represented as tensor of probabilities as float values in [0, 1] range of the size [256x256x6] and [512, 512, 6] where 6 is the number of predicted classes. The output will be post-processed (maxed) via mediapipe image segmenter API to get category mask.

LICENSED UNDER

Apache License, Version 2.0

Intended Uses

APPLICATIONS

- Human segmentation from images/videos in interactive entertainment, and video conferencing applications.

DOMAIN & USERS

- Augmented Reality
- Video Conferencing
- Entertainment

OUT-OF-SCOPE APPLICATIONS

Not appropriate for:

- This model is not intended for human life-critical decisions and applications that require pixel perfect masks
- Any form of surveillance or identity recognition is explicitly out of scope and not enabled by this technology.

Limitations

TRADE-OFFS

The model is optimized for real-time performance in the web browser and on a wide variety of mobile devices, and may not provide pixel perfect masks.

ENVIRONMENT

When degrading the environment light, backlit subjects, adding noise, or fast motions, or including large occluders, one can expect degradation of quality of the predicted mask.

Factors and Subgroups

ENVIRONMENTS

- Model is trained on thousands of samples with diverse identities (i.e. unique subjects), various blendshape combinations representing both common and random expressions and transformations (e.g. rotations).

Evaluation, Datasets and Results

Skin Tone and Gender Evaluation

DATASET

Contains 918 images, totalling 1902 instances of human subjects. Some images contained groups of people. Each human subject was annotated with perceived gender (male and female) and skin tone (from 1 to 10) based on the monk scale.

FAIRNESS CRITERIA

The SDM of each subgroup should be within one standard deviation from those values of the entire dataset to be considered fair.

EVALUATION RESULTS

Detailed evaluation across genders and skin tones is presented in the tables below.

FAIRNESS RESULTS

Examine whether subgroups are within one standard deviation away from the metrics of the entire dataset:
Across Gender:

- SDM: worst case 80.89, difference is 0.2, within the standard deviation of 10.2

Across skin tone:

- Mean IoU worst case 71.86, difference is 9.24, within the standard deviation of 10.2

Observed discrepancy across different genders and skin tones is less than one defined in our fairness criteria. We therefore consider the model performing well across groups.

512x512 resolution model, The IoU was computed at the input image resolution.

Gender	Test subset items and %		mean IOU	STDEV
Male	759	39.9%	81.48	10.09
Female	1143	60.1%	80.89	10.50
Total	1902	100%	81.10	10.2

Gender evaluation, Signed Deviation

Skintone	Test subset items and %		mean IOU	STDEV
Tone_1 + Tone_2	819	43%	*81.70*	*10.23*
Tone_3	396	20%	*80.93*	*9.20*
Tone_4	129	6%	*82.17*	*9.64*
Tone_5	81	4%	*81.53*	*9.03*
Tone_6	168	8%	*81.04*	*11.22*
Tone_7	230	12%	*79.98*	*11.17*
Tone_8	48	2%	*80.45*	*9.85*
Tone_9 + Tone_10	31	1.6%	*71.86*	*14.32*
Total	1902	100%	81.10	10.2

Skin tone evaluation, Signed Deviation

256x256 resolution model, The IoU was computed at the input image resolution.

Gender	Test subset items and %		mean IOU	STDEV
Male	759	39.9%	77.10	11.46
Female	1143	60.1%	77.32	10.94
Total	1902	100%	77.23	11.14

Gender evaluation, Signed Deviation

Skintone	Test subset items and %		mean IOU	STDEV
Tone_1 + Tone_2	819	43%	77.22	10.99
Tone_3	396	20%	77.59	11.37
Tone_4	129	6%	78.81	10.60
Tone_5	81	4%	78.16	10.27
Tone_6	168	8%	79.18	10.69
Tone_7	230	12%	75.98	11.08
Tone_8	48	2%	73.46	11.93
Tone_9 + Tone_10	31	1.6%	68.25	12.52
Total	1902	100%	77.23	11.14

Skin tone evaluation, Signed Deviation

Definitions

`AUGMENTED REALITY (AR)`
Augmented reality, a technology that superimposes a computer-generated image on a user's view of the real world, thus providing a composite view.

`INTERSECTION OVER UNION`
A measure of similarity. In the segmentation case, the ratio between the area of intersection of two masks and the area covered by their union.

Appendix

`List of predicted classes`

1 - background
2 - hair
3 - body-skin
4 - face-skin
5 - clothes
6 - others (accessories)